



Computational Chemistry

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- I. History of Computational Chemistry
- II. Methods Used in Computational Chemistry
- III. Applications of Computational Chemistry
- IV. Outlook for Computational Chemistry

GLOSSARY

ab initio Latin term meaning *from the beginning*. In the context of computational chemistry, computations from first principle without any empirical input except fundamental physical constants.

Density functional theory (DFT) Theory based on the electron density as the crucial property rather than the wave function in traditional *ab initio* methods.

Force field A set of equations describing the potential energy surface of a chemical system.

Hamiltonian operator An operator that describes the kinetic and potential energy of a system treated by wave mechanics.

Molecular mechanics Theoretical treatment of molecules by a force field based on classical mechanics and electrostatics.

Molecular modeling Branch of computational chemistry concerned with computer-aided molecular design.

Orbital Function to describe a single electron. Molecular orbitals (MOs) build the total wave function of a system and are expanded in terms of atomic orbitals, AOs (basis functions). Orbitals can be occupied or virtual.

Quantum mechanics Mathematical treatment based on the wavelike nature of small particles.

Schrödinger equation A differential equation for the quantum-mechanical treatment of a system.

Self-consistent field (SCF) method Method used to solve mathematical equations which depend on their own solution.

Semiempirical Making use of experimental results to derive parameters for approximations made in quantum-mechanical methods.

COMPUTATIONAL CHEMISTRY is the scientific discipline of applying computers to gain chemical information. It is the link between theoretical and experimental chemistry. Theoretical chemistry is mainly concerned with the development of mathematical models which allow one to derive chemical properties from calculations and to interpret experimental observations. The mathematical models developed in theoretical chemistry are usually validated by comparison with experiment. Theoretical chemistry existed before the arrival of electronic computers. Computational chemistry, however, relies heavily on powerful microelectronics to cope with huge computational tasks. It focuses on the application of theoretical methods which require calculational treatments which are by far too large to be done without fast computers.